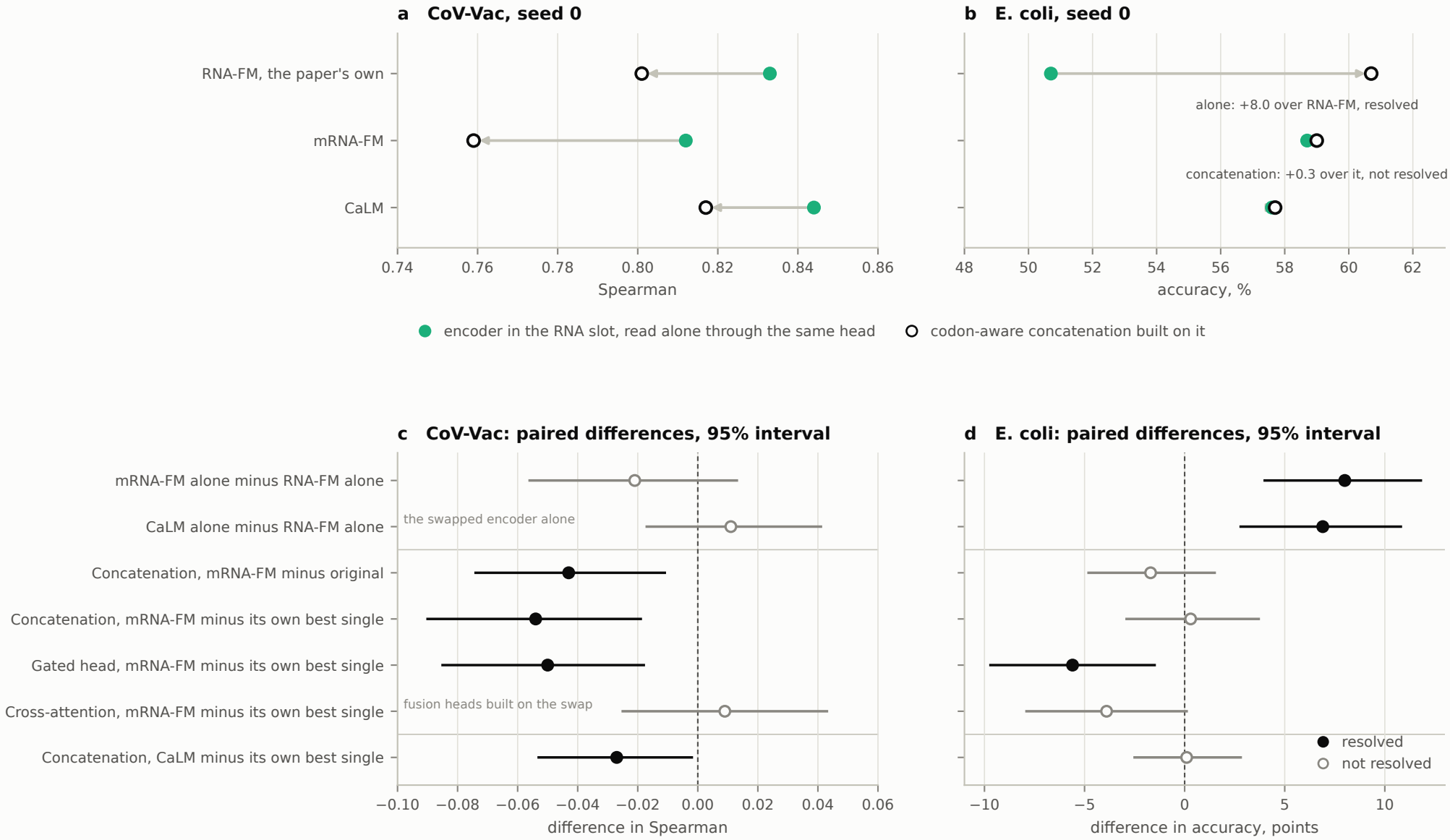
a CoV-Vac, seed 0
RNA-FM, the paper's own
mRNA-FM
CaLM
0.74
0.76
0.78
0.80
0.82
0.84
0.86
Spearman
b E. coli, seed 0
alone: +8.0 over RNA-FM, resolved
concatenation: +0.3 over it, not resolved
48
50
52
54
56
58
60
62
accuracy, %
encoder in the RNA slot, read alone through the same head
codon-aware concatenation built on it
c CoV-Vac: paired differences, 95% interval
mRNA-FM alone minus RNA-FM alone
CaLM alone minus RNA-FM alone
the swapped encoder alone
Concatenation, mRNA-FM minus original
Concatenation, mRNA-FM minus its own best single
Gated head, mRNA-FM minus its own best single
Cross-attention, mRNA-FM minus its own best single
fusion heads built on the swap
Concatenation, CaLM minus its own best single
−0.10
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
difference in Spearman
d E. coli: paired differences, 95% interval
resolved
not resolved
−10
−5
0
5
10
difference in accuracy, points